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2.1 Data Representation
2.2 Statistical Data Analysis
2.3 Data Visualization
2.4 Dimensionality Reduction
Video: Dimensionality Reduction 55 min
Graded App Item: How many dimensions are needed? Submitted
Graded App Item: Plotting the ellipse fit to the data Started
Quiz: PCA 15 min
2.5 Data Classification
2.6 Data Prediction

● Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

PCA

To pass 50% or higher

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Review Learning Objectives

1. In the attached '.mat' file you will find a data matrix 'dataset', containing 491 samples of a 10 feature dataset. This 10-dimensional dataset contains a "secret" 2D message, that you will find difficult, if not impossible, to decode with standard scatter plots of combinations of different dimensions. The 2D subspace where the message is legible corresponds to the two directions of greatest variance in the dataset. Thus, principal components analysis can be used to decode the message.
Due Apr 21, 11:59 PM IST
Try again

• Use the 'pca' function to perform principal components analysis.

• The PCA coordinates for the datapoints are contained in the second output of 'pca'. This is the representation of the data in the PCA linear vector space, where the first dimension is the direction of greatest variance, the 2nd dimension is the direction orthogonal to the first that has the greatest variance, and so on.

What is the maximum value of the data in the first dimension of the PCA linear vector space, rounded to the nearest integer?

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DimensionReduction

MAT File

120

12

363

Correct

2. What is the maximum value of the data in the second dimension of the PCA linear vector space, rounded to the nearest integer?

1 / 1 point

12

120

363

Correct

3. Scatter plot the coordinates of the samples in the PCA linear vector space in the first two PCA dimensions (the PCA coordinates for the datapoints are contained in the second output of 'pca'). This represents the data in a 2-D reduced space of the 10-D dataset. To ensure it is displayed in the proper aspect ratio, use:

1 / 1 point

1 axis = equal;

What is the secret message?

exp(i*pi)=1+0

Correct

https://www.coursera.org/learn/matrix-image-processing/exam/07bb1/pcskltemg7nedirectToCover>true

1/1